Nome: Matheus Santos de Oliveira

Cartão: 00326048

# Tarefa 10

**Figura 1 – Sinal original no tempo**

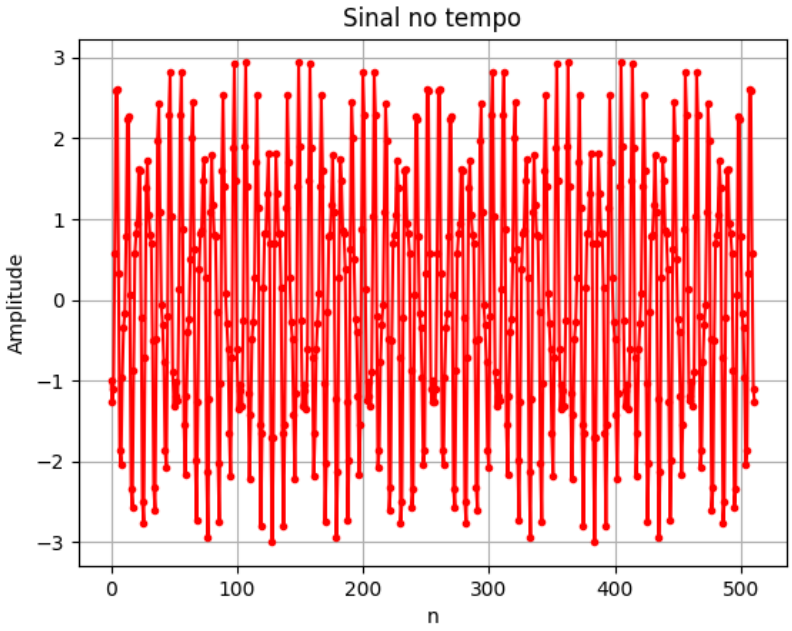



**Figura 2 – FFT do sinal original**

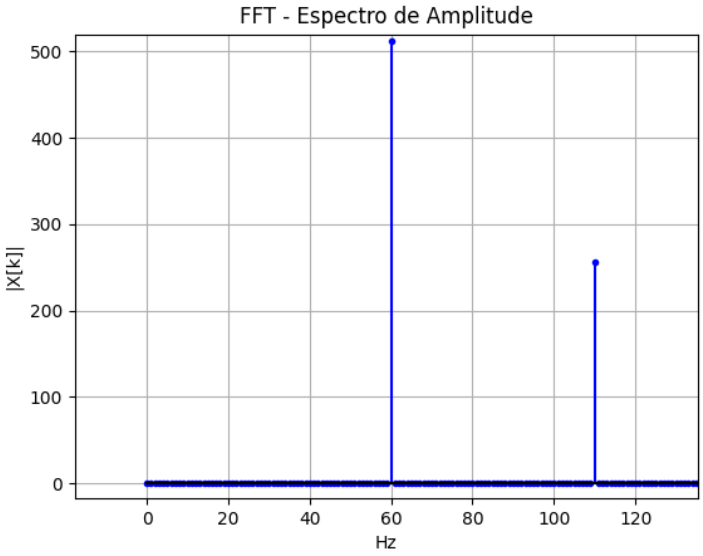


**Figura 3 – Sinal com ruído no tempo**

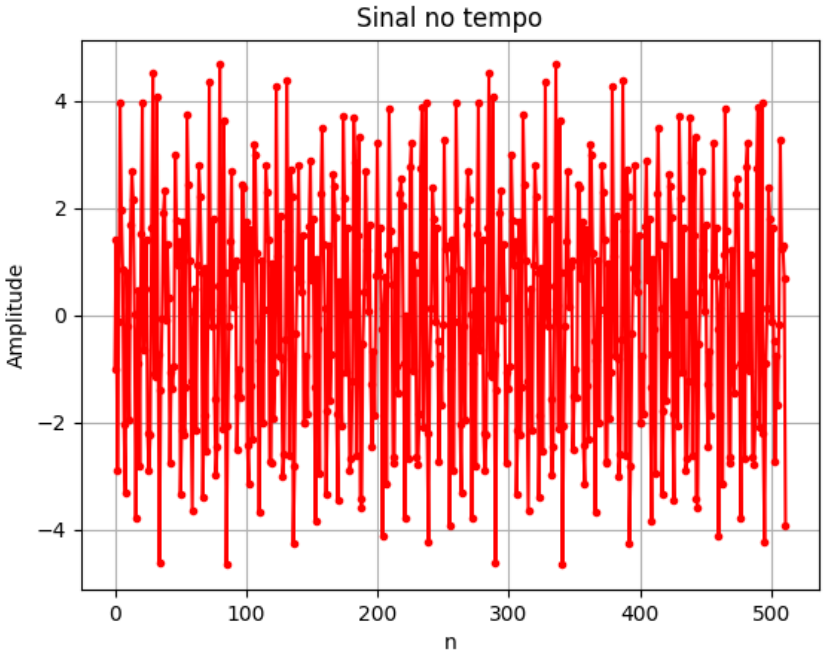
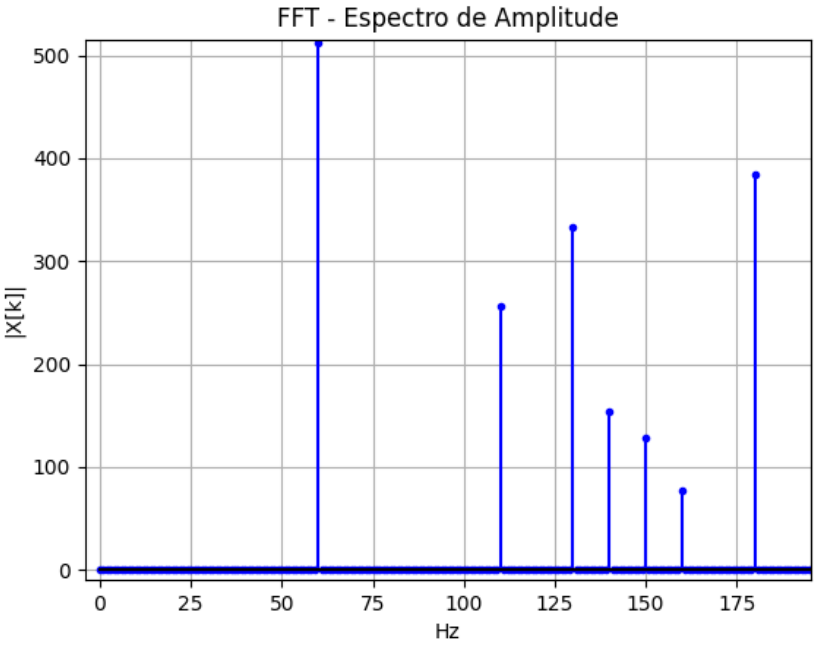



**Figura 4 – FFT do sinal com ruído**



## ***<u>Caso 1</u>: Filtro Butterworth***

**Figura 5 – Sinal filtrado no tempo**

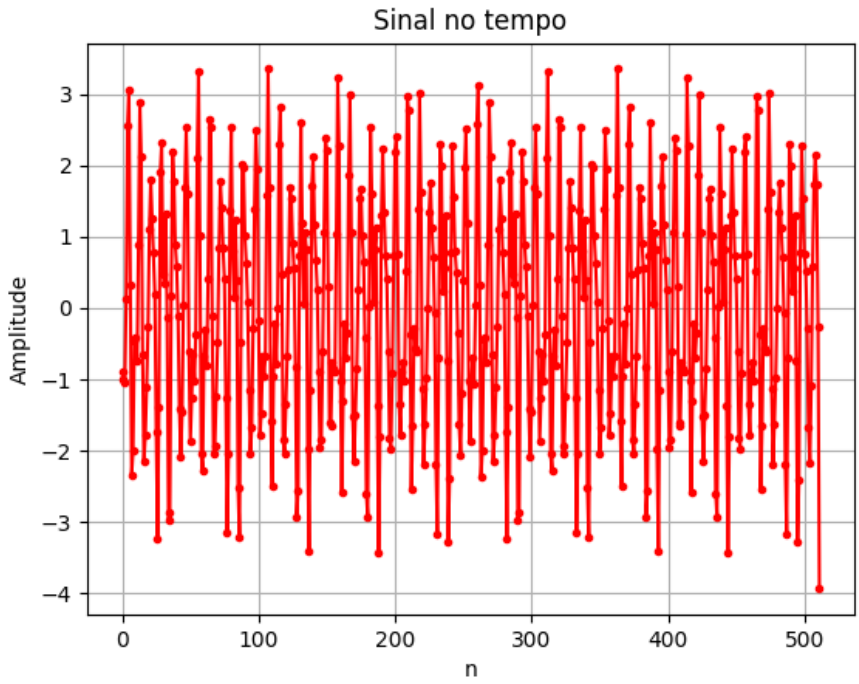



**Figura 6 – FFT do sinal filtrado**

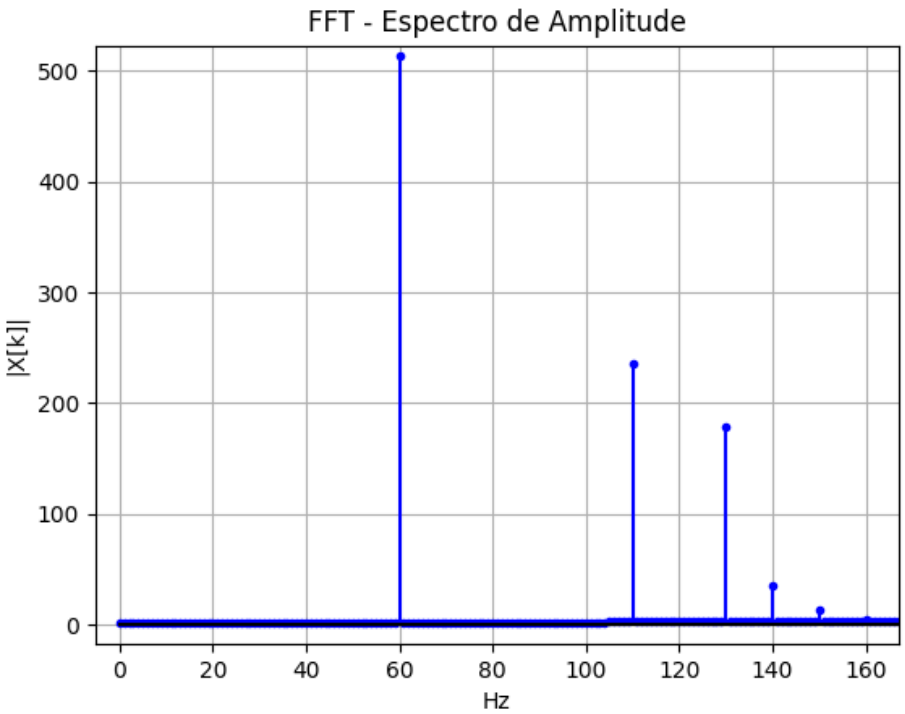
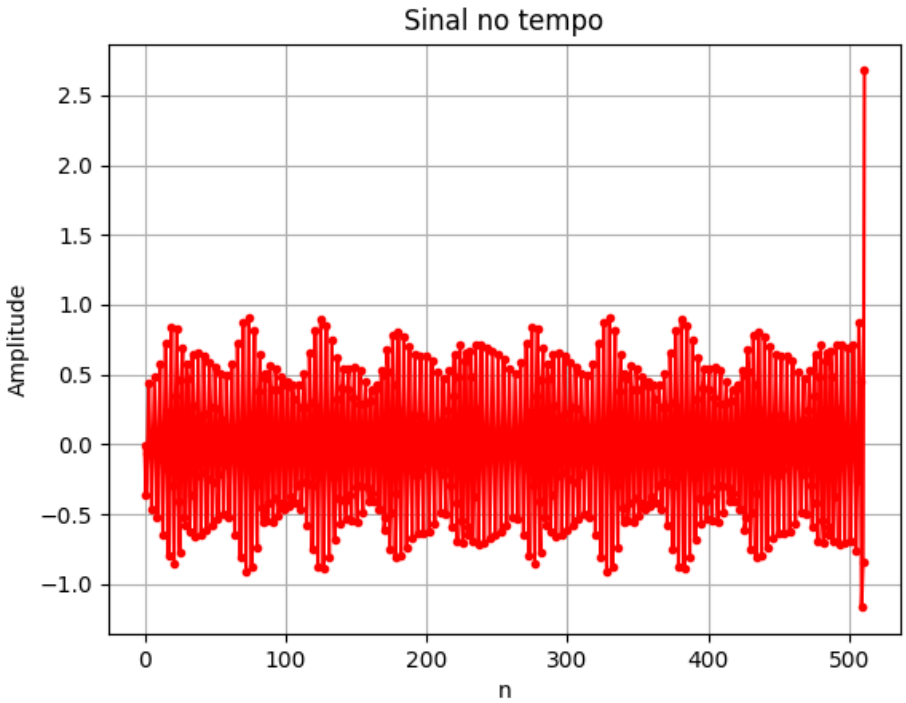


**Figura 7 – Erro entre o sinal original e o sinal filtrado**




**Erro Quadrático Médio (MSE): 2.697800e-01**

**N (ordem) = 5**

**Fc ≈ 131 Hz**

**Coeficientes do numerador (b):**

**[0.0578102 0.289051 0.57810201 0.57810201 0.289051 0.0578102 ]**

**Coeficientes do denominador (a):**

**[1. 0.11525048 0.63808188 0.03913043 0.05643495 0.00102869]**

***<u>Caso 2</u>: Filtro janelado (Hamming, M=64), com fc igual ao filtro de Butterworth***

**Figura 8 – Sinal filtrado no tempo**

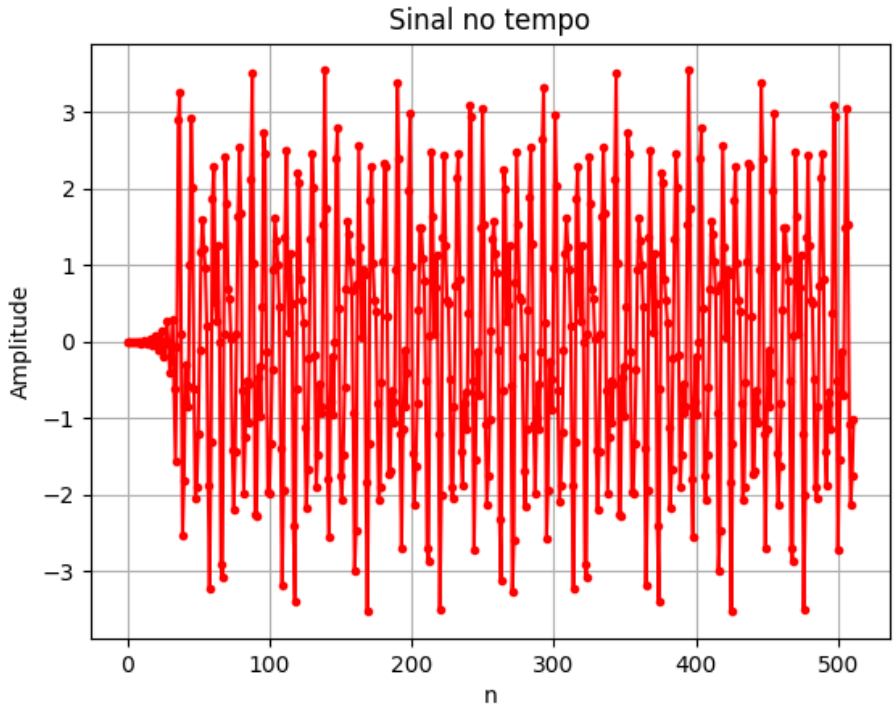



**Figura 9 – FFT do sinal filtrado**

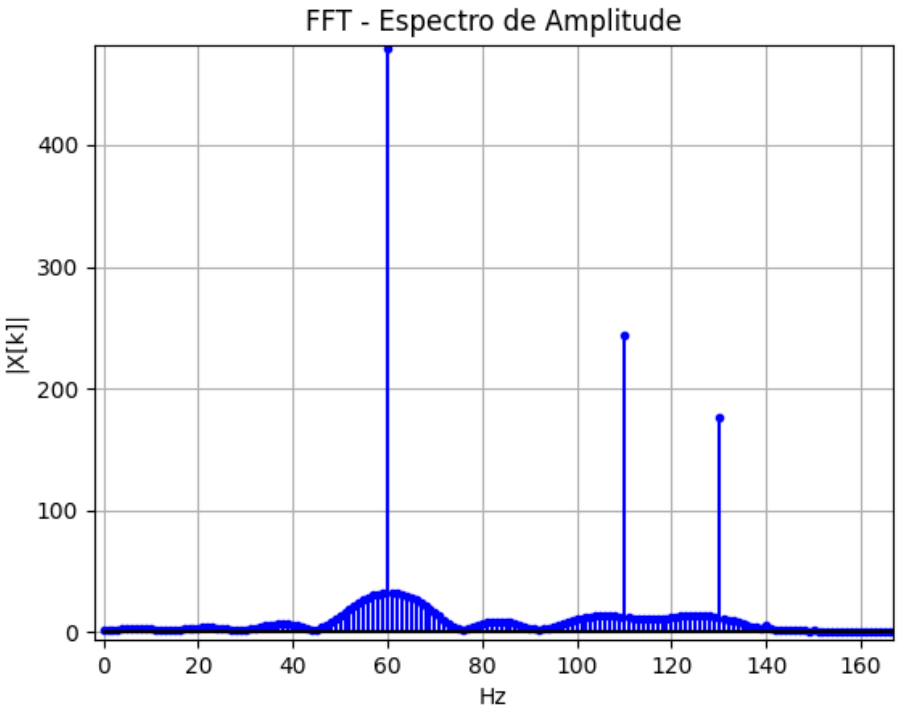
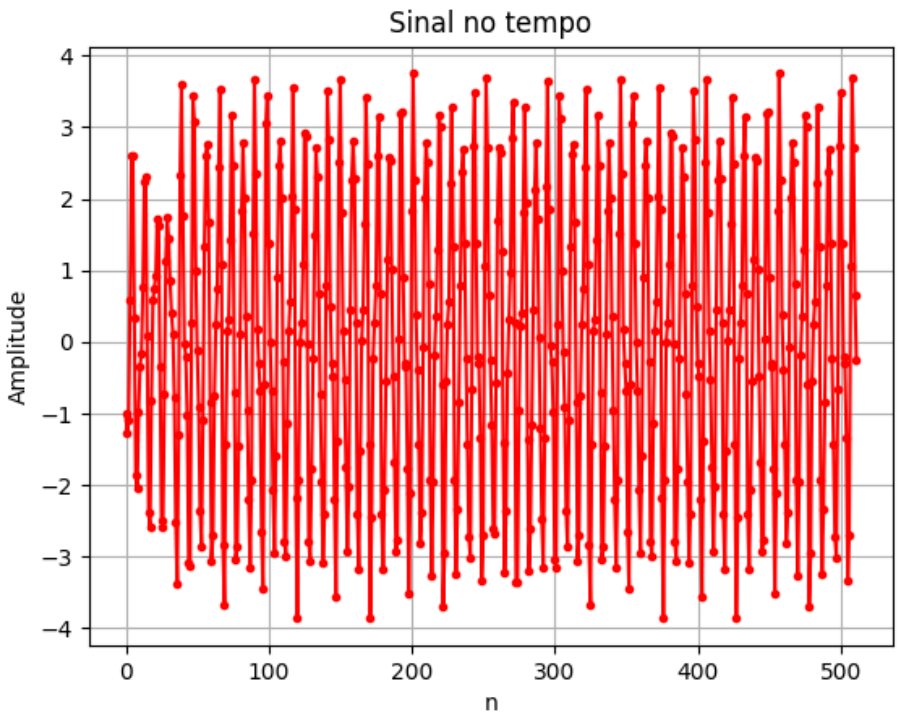


**Figura 10 – Erro entre o sinal original e o sinal filtrado**




**Erro Quadrático Médio (MSE): 3.281259e+00**

## <u>Conclusão</u>:

O filtro Butterworth atendeu fielmente às especificações, produzindo uma zona de transição suave entre 120 e 200 Hz, sem ripple perceptível e com o menor MSE entre os dois métodos analisados. Já o filtro FIR janelado com Hamming, embora tenha apresentado maior ripple e um MSE superior, demonstrou transição mais abrupta, alcançando a banda de rejeição mais cedo do que o Butterworth. Assim, enquanto o Butterworth se destacou pela menor distorção e maior fidelidade ao sinal, o FIR apresentou melhor seletividade, com rejeição antecipada.